

CO5_AT3_SIMATS Browser Level 3 Exam

Level Exam </> Java

← Back

QUESTIONS

LEVEL 3 — HARD

Q1

Numbers

Q2

Control

2 / 2 saved

Level 3 — Hard (H) Factors — Numbers

Write a program to print number of factors and to print nth factor of the given number.

Sample Input:

Given Number: 100

N = 4

Sample Output:

Number of factors = 9

4th factor of 100 = 5

Testcase: 1. Given Number = 512, N = 6

2. Given Number = 343, N = 7

3. Given Number = 1024, N = 0

4. Given Number = -6561, N = 3

5. Given Number = 0, N = 2

Your Code

```
1 import java.util.Scanner;  
2  
3 public class Main {  
4     public static void main(String[] args) {  
5         Scanner sc = new Scanner(System.in);  
6         int num = sc.nextInt();  
7         int n = sc.nextInt();
```

Program Input (stdin)

Enter input if required...

Output

Number of Factors = 9

CO5_AT3_SIMATS Browser Level 3 Exam

Level Exam </> Java

← Back

QUESTIONS

LEVEL 3 — HARD

Q1

Numbers

Q2

Control

2 / 2 saved

Level 3 — Hard (H) Steps — Control

Given an integer num, return the number of steps to reduce it to zero. In one step, if the current number is even, you have to divide it by 2, otherwise, you have to subtract 1 from it.

Example 1:

Input: num = 14

Output: 6

Explanation:

Step 1) 14 is even; divide by 2 and obtain 7.

Step 2) 7 is odd; subtract 1 and obtain 6.

Step 3) 6 is even; divide by 2 and obtain 3.

Step 4) 3 is odd; subtract 1 and obtain 2.

Step 5) 2 is even; divide by 2 and obtain 1.

Step 6) 1 is odd; subtract 1 and obtain 0.

Testcase:1. n = 5

2. n = 10

3. n = 12

4. n = 18

5. n = 20

Your Code

```
1 import java.util.Scanner;  
2  
3 public class Main {
```

Program Input (stdin)

Enter input if required...